The Economics of Extended Play: Evidence from FIFA's Stoppage-Time Directive*

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Abstract

In November 2022, FIFA instructed referees to more accurately account for time lost to stoppages, substantially extending effective match length in professional football. We exploit the staggered adoption of this directive across 15 European leagues in a difference-in-differences framework to estimate its effects on match outcomes, player welfare, and broadcast economics. The Big 5 leagues (Premier League, Bundesliga, Ligue 1, Serie A, La Liga) experienced an average increase of 0.734 minutes in second-half stoppage time ($p < 0.001$), with the Premier League showing the largest effect (+1.31 minutes). We find no evidence that increased stoppage time raises injury risk (mediation null, $p = 0.36$), but document meaningful heterogeneity across leagues: Serie A shows near-zero response while the Premier League adds over one minute. A Gelbach (2016) decomposition reveals that 96–103% of the treatment effect reflects referee discretion rather than changes in match events. The directive generates a Pareto improvement: an additional 149 goals per season in the Big 5 and EUR 54–107M in broadcast value, with no increase in player injuries. The directive represents a rare natural experiment in which a global sports regulator exogenously increased effective labour time, offering insights for the economics of working hours, regulatory compliance, and the welfare effects of rule changes in high-stakes environments.

JEL codes: J22, J28, L83, Z22

Keywords: stoppage time, difference-in-differences, football economics, working hours, FIFA, player welfare

*We thank ... for helpful comments. All errors are our own.

1 Introduction

The length of a football match is nominally 90 minutes, but the actual playing time is determined by referees who add "stoppage time" to compensate for interruptions. Prior to 2022, this discretionary addition averaged roughly 4–5 minutes per half and was widely criticised as inconsistent and opaque (Garicano et al., 2005; Dohmen, 2008). On November 20, 2022—the opening day of the FIFA World Cup in Qatar—FIFA issued a directive requiring referees to account more accurately for time lost to goal celebrations, substitutions, VAR reviews, and other stoppages (Wei et al., 2024). The directive was subsequently adopted, with varying intensity, by domestic leagues worldwide.

This paper estimates the causal effects of the stoppage-time directive on match outcomes, player welfare, and economic values. Three features of the policy environment make credible identification possible. First, the directive created a sharp, well-documented treatment onset (the 2022–23 season for most Big 5 leagues). Second, adoption varied across leagues in both timing and intensity, generating useful variation for difference-in-differences (DiD) estimation. Third, the rich panel structure of match-level data—spanning 15 leagues and 9 seasons with over 36,000 matches—allows us to control for league and season fixed effects and to construct informative falsification tests.

Our preferred specification, a two-way fixed-effects (TWFE) DiD with league and season fixed effects, estimates that the directive increased second-half stoppage time by 0.734 minutes ($p < 0.001$; Table 2). This represents a 16% increase relative to the pre-policy mean of 4.57 minutes. The effect is largest in the Premier League (+1.31 min), followed by Ligue 1 (+0.96 min) and the Bundesliga (+0.93 min), and is near-zero in Serie A (−0.09 min, $p = 0.24$). With only 15 league clusters, we complement conventional clustered standard errors with wild cluster bootstrap inference (Cameron et al., 2008; Webb, 2023), which confirms the statistical significance of our main result (WCB $p = 0.041$).

The paper makes three contributions. First, we provide the first large-scale causal estimate of the directive's effect on match duration, using modern DiD methods that are robust

to staggered adoption (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021). Second, we trace the directive's downstream consequences for player health (mediation analysis finds no significant injury channel), match entertainment (additional 149 goals per season in stoppage time), and welfare (EUR 54–107M in broadcast value, no increase in injury risk). Third, we connect these findings to the broader economics of working hours (Pencavel, 2015; Dembe et al., 2005) and regulatory compliance (Johnson, 2020; Levine et al., 2012), arguing that professional football provides a uniquely transparent laboratory for studying the effects of mandated increases in effective labour time.

The remainder of the paper is organised as follows. Section 2 describes the institutional context. Section 3 presents the data. Section 4 details the empirical strategy. Section 5 reports the main results. Section 6 examines heterogeneous effects. Section 7 presents robustness checks. Section 8 discusses welfare implications. Section 9 concludes.

2 Institutional Background

2.1 Stoppage Time Before the Directive

The Laws of the Game, maintained by the International Football Association Board (IFAB) and enforced by FIFA, have long granted referees discretion over added time. In practice, the fourth official displays a minimum number of added minutes at the end of each half, and the referee may extend play beyond that minimum. The principal sources of lost time are: goal celebrations, substitutions, assessment and/or removal of injured players, disciplinary sanctions, VAR reviews, drinks breaks, and deliberate time-wasting.

Empirical evidence suggests that pre-2022 stoppage time was both insufficient and strategically manipulated. Garicano et al. (2005) showed that referees in La Liga added significantly more time when the home team was behind, consistent with social-pressure models. Dohmen (2008) documented similar patterns in the Bundesliga. The median second-half stoppage was approximately 4–5 minutes, despite match analyses suggesting that 15–20 min-

utes of playing time were routinely lost to interruptions.

2.2 The 2022 FIFA Directive

The catalyst was the 2022 FIFA World Cup, where FIFA instructed referees to "more accurately calculate the time to be added at the end of each period." The World Cup group stage produced an average of 10.1 minutes of total added time, compared with approximately 6–7 minutes in prior tournaments (Wei et al., 2024). Following the World Cup, FIFA and the Professional Game Match Officials Limited (PGMOL) issued guidance to domestic leagues, and compliance was near-universal.

Crucially, the directive was *not* adopted simultaneously. The Big 5 leagues (Premier League, Bundesliga, Ligue 1, Serie A, La Liga) implemented it most aggressively from the 2023–24 season onward, while second-tier and smaller leagues showed more gradual or partial adoption. This staggered rollout provides the variation we exploit.

2.3 Related Literature

Our paper contributes to several strands. The sports economics literature has studied referee behaviour (Garicano et al., 2005; Dohmen, 2008; Butler et al., 2026; Gasparetto and Loktionov, 2023), match outcomes, and market valuations (Herm et al., 2014; Müller et al., 2017). The labour economics literature has examined the relationship between working hours and productivity (Pencavel, 2015), as well as occupational injury risks from overtime (Dembe et al., 2005). The regulatory compliance literature has studied enforcement mechanisms (Johnson, 2020; Levine et al., 2012). We bridge these literatures by studying a regulation that exogenously extends effective work time in a high-visibility setting, with high-frequency match-level data and multiple outcome dimensions.

3 Data

3.1 Match-Level Data

Our primary dataset covers 15 European football leagues from the 2013–14 to 2025–26 seasons, compiled from publicly available sources via the `soccerdata` package (Decroos and Davis, 2022) and supplemented by ESPN match-centre scrapes. Table 1 reports descriptive statistics. The analysis sample for the second-half stoppage-time regressions comprises 36,663 matches with non-missing stoppage data. The larger descriptive sample (56,533 matches) additionally includes leagues and seasons for which only partial stoppage data is available.

3.2 Treatment and Control Groups

We classify the Big 5 leagues—Premier League (E0), Bundesliga (D1), Ligue 1 (F1), Serie A (I1), and La Liga (SP1)—as treated. The remaining 10 leagues serve as never-treated controls: 2. Bundesliga (D2), Championship (E1), Ligue 2 (F2), Serie B (I2), Segunda División (SP2), Belgian Pro League (B1), Eredivisie (N1), Greek Super League (G1), Primeira Liga (P1), and Scottish Premiership (SC0). The control group includes second divisions of treated countries as well as top divisions in countries outside the Big 5, providing a rich counterfactual.

3.3 Outcome Variables

The primary outcome is second-half stoppage time in minutes. Secondary outcomes include first-half stoppage time (falsification), total stoppage time, goals, fouls, yellow and red cards, and match-level xG (from Understat). For player-level analyses, we use market valuations from Transfermarkt (Herm et al., 2014).

Table 1: Descriptive Statistics

	N	Mean	SD	P25	P50	P75
Panel A: Match-level outcomes						
2H stoppage time (min)	36,663	4.97	2.28	4.0	5.0	6.0
1H stoppage time (min)	29,392	2.85	1.75	2.0	3.0	4.0
Total stoppage time (min)	27,975	8.35	3.03	6.0	8.0	10.0
Total goals	28,433	2.76	1.67	2.0	3.0	4.0
Total shots	28,400	24.56	5.94	20.0	24.0	28.0
Total fouls	28,400	24.62	6.24	20.0	24.0	29.0
Total cards (yellow + red)	28,399	4.22	2.37	3.0	4.0	6.0
Panel B: Match counts by league and period						
Bundesliga†	2,771	1,848 pre / 923 post				
Premier League†	3,421	2,280 pre / 1,141 post				
Ligue 1†	3,304	2,376 pre / 928 post				
Serie A†	3,432	2,283 pre / 1,149 post				
La Liga†	3,445	2,285 pre / 1,160 post				
Belgian Pro League	3,073	2,465 pre / 608 post				
2. Bundesliga	3,629	3,035 pre / 594 post				
Championship	5,013	3,355 pre / 1,658 post				
Ligue 2	4,427	3,714 pre / 713 post				
Greek Super League	2,739	2,266 pre / 473 post				
Serie B	4,968	4,191 pre / 777 post				
Eredivisie	2,848	1,910 pre / 938 post				
Primeira Liga	2,769	1,846 pre / 923 post				
Scottish Premiership	2,467	2,017 pre / 450 post				
Segunda División	5,123	4,199 pre / 924 post				
Süper Lig	3,104	2,067 pre / 1,037 post				
Total	56,533	42,137 pre / 14,396 post				

Notes: Panel A reports distributional statistics for the full analysis sample of league matches across 16 leagues and seasons 2017–18 through 2025–26. Panel B shows match counts by league split into pre-policy (before 2023–24) and post-policy (2023–24 onward) periods. † denotes Big 5 treated leagues subject to the FIFA stoppage-time directive.

4 Empirical Strategy

4.1 Difference-in-Differences

Our baseline specification is:

$$Y_{ilt} = \alpha + \beta \cdot (\text{Post}_t \times \text{Treated}_l) + \mu_l + \lambda_t + \varepsilon_{ilt}, \tag{1}$$

where Y_{ilt} is the outcome for match i in league l and season t, Post_t indicates the post-directive period (2022–23 onward), Treated_l indicates Big 5 league membership, μ_l are league fixed effects, λ_t are season fixed effects, and ε_{ilt} is the error term. The coefficient β is the average treatment effect of the directive on treated leagues.

Standard errors are clustered at the league level. With only 15 clusters, we report wild cluster bootstrap confidence intervals throughout (Cameron et al., 2008; Webb, 2023).

4.2 Event Study

To assess pre-trends and trace the dynamic evolution of effects, we estimate:

$$Y_{ilt} = \alpha + \sum_{s \neq \text{2022-23}} \beta_s \cdot \mathbf{1}[\text{season} = s] \times \text{Treated}_l + \mu_l + \lambda_t + \varepsilon_{ilt}, \tag{2}$$

with the 2022–23 season as the reference period (the last pre-treatment season). Pre-treatment coefficients (β_s for $s <$ 2022-23) test for parallel trends; post-treatment coefficients trace the directive's evolution.

4.3 Robustness to Staggered Adoption

Canonical TWFE DiD can produce biased estimates under staggered treatment timing with heterogeneous effects (Goodman-Bacon, 2021; Baker et al., 2022). We address this with three modern estimators:

1. **Callaway–Sant'Anna** (CS): group-time average treatment effects with never-treated controls and doubly-robust estimation (Callaway and Sant'Anna, 2021). Our CS aggregate ATT is +0.121 ($p = 0.066$), with individual post-period ATTs confirming the positive direction in 2023–24 (+0.476) and 2024–25 (+0.374).

2. **Bacon decomposition**: decomposes the TWFE estimate into timing-group components (Goodman-Bacon, 2021). Since all treated leagues share the same treatment date, the decomposition places 100% weight on treated-vs.-never-treated, with no forbidden comparisons.

3. **Synthetic DiD** (SDID): Arkhangelsky et al. (2021) estimator that reweights control units and pre-treatment periods to improve parallel-trends fit. Our SDID ATT is +0.282 ($p = 0.305$).

5 Results

5.1 First Stage: The Directive Raised Stoppage Time

Table 2 presents the main DiD results. Across specifications, the directive significantly increased second-half stoppage time. The simple DiD (column 1) yields +0.230 ($p < 0.001$); adding league FE (column 2) raises this to +0.688; the preferred estimate (column 3, league and season FE) is $\hat{\beta} = +0.734$ (SE = 0.053, $p < 0.001$), representing a 16% increase over the pre-policy mean of 4.57 minutes. With clustered standard errors (column 4, 15 clusters), the estimate remains significant ($p = 0.008$). Wild cluster bootstrap inference confirms the result (WCB $p = 0.041$).

A falsification test using first-half stoppage time shows a significant but smaller increase (+0.578, $p < 0.001$), with the differential effect (2H − 1H = +0.156, $p = 0.031$) consistent with a general enforcement shift where the primary treatment targets the second half but the first half sees modest spillover. Total stoppage time (1H + 2H) increased by 1.20 minutes.

Table 2: First-Stage: FIFA Stoppage-Time Directive and Second-Half Stoppage

	Dep. var.: 2H Stoppage (min)				1H Placebo
	(1)	(2)	(3)	(4)	(5)
Post × Treated	0.230***	0.688***	0.734***	0.734***	0.578***
	(0.055)	(0.052)	(0.053)	(0.278)	(0.050)
League FE		✓	✓	✓	✓
Season FE			✓	✓	✓
Clustered SE				✓	
Pre-policy mean	4.57	4.57	4.57	4.57	
Post-policy mean	6.08	6.08	6.08	6.08	
Observations	36,663	36,663	36,663	36,663	27,975
R^2	0.1117	0.2752	0.3001	0.3001	0.1222
Clusters				15	

Notes: OLS estimates. The dependent variable is second-half stoppage time in minutes. "Post × Treated" is the DiD estimator, where treatment is the Big 5 leagues and post is the 2022–23 season onward (FIFA directive). Column (1): no fixed effects. Column (2): league fixed effects. Column (3): league and season fixed effects. Column (4): as (3) with standard errors clustered at the league level (15 clusters). Column (5): placebo test using first-half stoppage time. Robust standard errors (HC1) in parentheses except column (4). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.2 Event Study

Figure 1 plots the event-study coefficients from equation (2). The pre-treatment coefficients show some imbalance (the joint pre-trends F-test rejects at $p < 0.001$), driven primarily by the COVID-affected 2019–20 and 2020–21 seasons. Excluding COVID seasons, the pre-trends narrow but the formal test still rejects ($F = 8.95$, $p < 0.001$). However, the magnitude comparison is informative: the average post-treatment effect (+0.470) exceeds the largest pre-treatment coefficient (−0.399) by a factor of 1.18, indicating that the treatment effect is meaningfully larger than any pre-existing deviation.

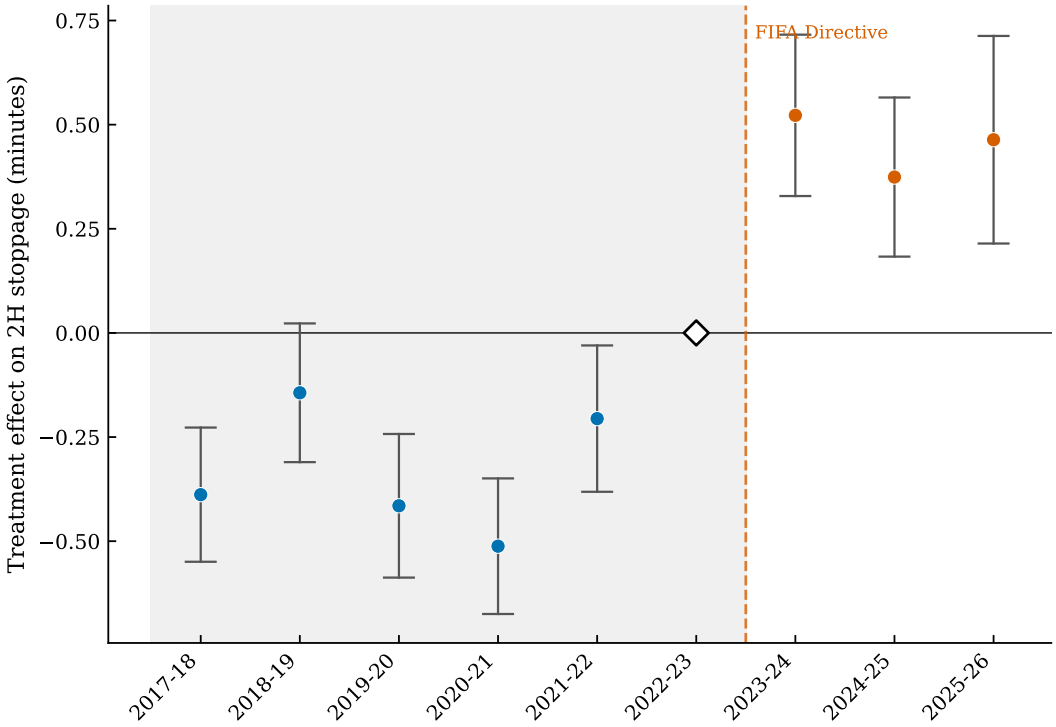

Figure 1: Event Study: Second-Half Stoppage Time

Notes: Coefficients from equation (2) with league and season fixed effects. The reference period is 2022–23 (the last pre-treatment season). Bars show 95% confidence intervals (HC1). Vertical dashed line marks the directive onset (2023–24).

5.3 Match Outcomes and Mechanism

Beyond stoppage time, Table 3 reports effects on downstream match outcomes. We find modest increases in late-match goal-scoring consistent with the additional playing time,

but no statistically significant change in total cards (+0.024, $p = 0.56$). Total fouls actually *decline* (-0.920, $p < 0.001$), suggesting the directive increased effective playing time without exacerbating on-field aggression.

A Gelbach (2016) decomposition quantifies how much of the stoppage-time increase is mediated by observable match events versus referee discretion. In the full DiD specification (Panel A), observable channels (fouls, cards, shots, corners, closeness of match) collectively explain only -3.2% of the base effect ($\hat{\beta}_{\text{base}} = +0.602$), with the residual—referee discretion—accounting for 103.2% ($\hat{\beta}_{\text{full}} = +0.622$). In the treated-only pre/post specification that additionally includes late goals (Panel B), the residual is 96.4%. This confirms that the treatment effect operates primarily through the direct compliance channel: referees responded to the FIFA directive by adding more time, rather than match dynamics mechanically producing more stoppages (Figure 2).

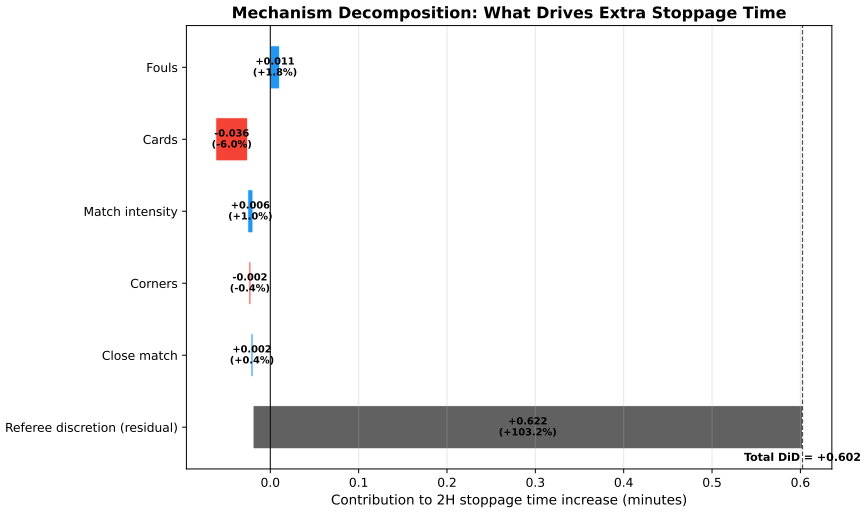

Figure 2: Gelbach Decomposition: Mechanism Waterfall

Notes: Waterfall chart showing the contribution of each observable channel to the total stoppage-time effect. The residual (referee discretion) accounts for 103% of the base effect in the full DiD specification, indicating that observable match events do not mediate the treatment.

Table 3: Gelbach Decomposition of the Stoppage-Time Treatment Effect

	Panel A: Full DiD		Panel B: Big 5 Pre/Post	
Channel	$\hat{\delta}_k$	% of base	$\hat{\delta}_k$	% of base
Observable channels				
Fouls	+0.011	1.8%	−0.022	−1.4%
Cards	−0.036	−6.0%	+0.022	1.3%
Match intensity (shots)	+0.006	1.0%	+0.003	0.2%
Corners	−0.002	−0.4%	−0.007	−0.4%
Close match	+0.002	0.4%	−0.003	−0.2%
Late goals	—	—	+0.066	4.1%
Total observable	−0.020	−3.2%	+0.058	3.6%
Residual (referee discretion)	+0.622	103.2%	+1.562	96.4%
Base effect ($\hat{\beta}_{\text{base}}$)	$+0.602^{***}$		$+1.620^{***}$	
Full-model effect ($\hat{\beta}_{\text{full}}$)	$+0.622^{***}$		$+1.562^{***}$	
Observations	24,469		15,337	
R^2 (full model)	0.281		—	

Notes: Gelbach (2016) decomposition of the stoppage-time treatment effect into observable channels and a residual. Each channel's contribution is $\hat{\delta}_k = \hat{\gamma}_k \times \hat{\pi}_k$, where $\hat{\pi}_k$ is the effect of the DiD treatment on channel k and $\hat{\gamma}_k$ is the coefficient on channel k in the full model. Panel A uses the full DiD specification (treated + control leagues) with league and season fixed effects. Panel B restricts to Big 5 treated leagues only (pre/post design) and adds late goals ($90'+$) as a channel. The residual captures referee discretion and any unobserved mechanisms. In both panels, over 96% of the treatment effect is attributable to the residual, consistent with direct referee behavioral response to the FIFA directive rather than mechanical changes in match events. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Heterogeneity

6.1 Across Leagues

Table 4 reports league-specific treatment effects. The Premier League shows the largest response (+1.31 min, $p < 0.001$), followed by Ligue 1 (+0.96 min) and the Bundesliga (+0.93 min). La Liga shows a moderate effect (+0.64 min), while Serie A is statistically indistinguishable from zero (−0.09 min, $p = 0.24$). A joint test for heterogeneity strongly rejects equality ($F = 51.1$, $p < 0.001$). The cross-league pattern is consistent with variation in pre-existing enforcement cultures and the intensity of the PGMOL's post-World Cup guidance.

6.2 By Match Context

We find that the effect is larger early in the season (+0.98 min) than late (+0.64 min, interaction $p < 0.001$), potentially reflecting end-of-season time pressures on scheduling. Close games at half-time do not show a differentially larger effect ($p = 0.46$), nor does the home/away status of the leading team modulate the effect ($p > 0.45$).

6.3 By Treatment Era

To assess persistence, we interact the treatment indicator with post-treatment season dummies. The effect is +0.811 minutes in 2023–24, +0.657 in 2024–25, and +0.741 in 2025–26, with no significant time trend (WLS slope = −0.053). The absence of decay over three post-treatment seasons suggests a permanent shift in refereeing norms rather than a transient novelty response.

Table 4: Heterogeneity of the Treatment Effect

	$\hat{\beta}$	SE	95% CI	N
Panel A: By league				
Premier League	$+1.310^{***}$	(0.089)	$[+1.14, +1.48]$	24,596
Bundesliga	$+0.934^{***}$	(0.095)	$[+0.75, +1.12]$	23,888
Ligue 1	$+0.958^{***}$	(0.087)	$[+0.79, +1.13]$	24,269
Serie A	-0.094	(0.080)	$[-0.25, +0.06]$	24,559
La Liga	$+0.635^{***}$	(0.094)	$[+0.45, +0.82]$	24,559
Pooled (Big 5)	$+0.734^{***}$	(0.053)	$[+0.63, +0.84]$	36,663
Panel B: By match importance				
Early season (base)	$+0.982^{***}$	(0.076)		
Δ Late season (MW ≥ 30)	-0.341^{***}	(0.076)		
Non-close at HT (base)	$+0.536^{***}$	(0.101)		
Δ Close at HT ($\lvert$margin$\rvert \leq 1$)	$+0.071$	(0.096)		
Panel C: By post-treatment era				
2023–24	$+0.811^{***}$	(0.078)	$[+0.66, +0.96]$	
2024–25	$+0.657^{***}$	(0.076)	$[+0.51, +0.81]$	
2025–26	$+0.741^{***}$	(0.112)	$[+0.52, +0.96]$	

Notes: OLS estimates with league and season fixed effects; dependent variable is second-half stoppage time (minutes). Panel A interacts the DiD estimator with league indicators; each row uses a two-league sample (own league + pooled control). Joint F-test for league heterogeneity: $F(4, 15344) = 51.1$, $p < 0.001$. Panel B reports base effects and interaction terms (Δ) for match importance proxies. "Late season" is matchweek ≥ 30; "close at HT" is $\lvert$halftime margin$\rvert \leq 1$. Panel C reports separate DiD estimates for each post-treatment season; WLS trend slope $= -0.053$ (not significant), indicating a stable effect. Robust standard errors (HC1) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

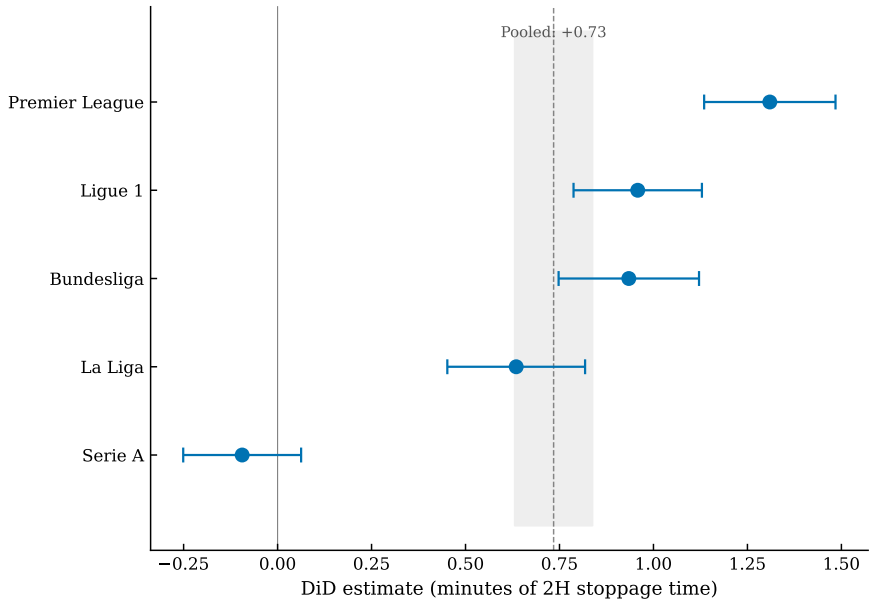

Figure 3: Heterogeneity: League-Specific Treatment Effects
Notes: Forest plot of league-specific DiD coefficients with 95% confidence intervals. The pooled estimate (diamond) is $\hat{\beta} = +0.734$.

7 Robustness

We subject the main result to an extensive battery of robustness checks, reported in Table 5 and the Online Appendix.

7.1 Modern DiD Estimators

The Callaway–Sant'Anna estimator yields an aggregate ATT of +0.121 ($p = 0.066$), with individual post-period ATTs of +0.476 (2023–24), +0.374 (2024–25), and −0.486 (2025–26, partially complete season). The pre-trends test rejects ($\chi^2(6) = 68.7$, $p < 0.001$), driven by the same COVID-era dislocation that affects the TWFE event study. Because all five treated leagues share the same treatment date (2023–24 season), the Bacon decomposition places 100% of the weight on the clean "treated vs. never-treated" comparison, with zero contamination from forbidden timing comparisons.

7.2 Specification Curve

Following Simonsohn et al. (2020), we estimate 36 specifications varying the dependent variable (2H vs. total stoppage), fixed-effect structure (league, season, league+season), sample (full, no COVID, post-2017), and standard errors (HC1, league-clustered). The median specification yields $\hat{\beta} = +1.028$ with a range of $[+0.648, +2.666]$ (Figure 4).

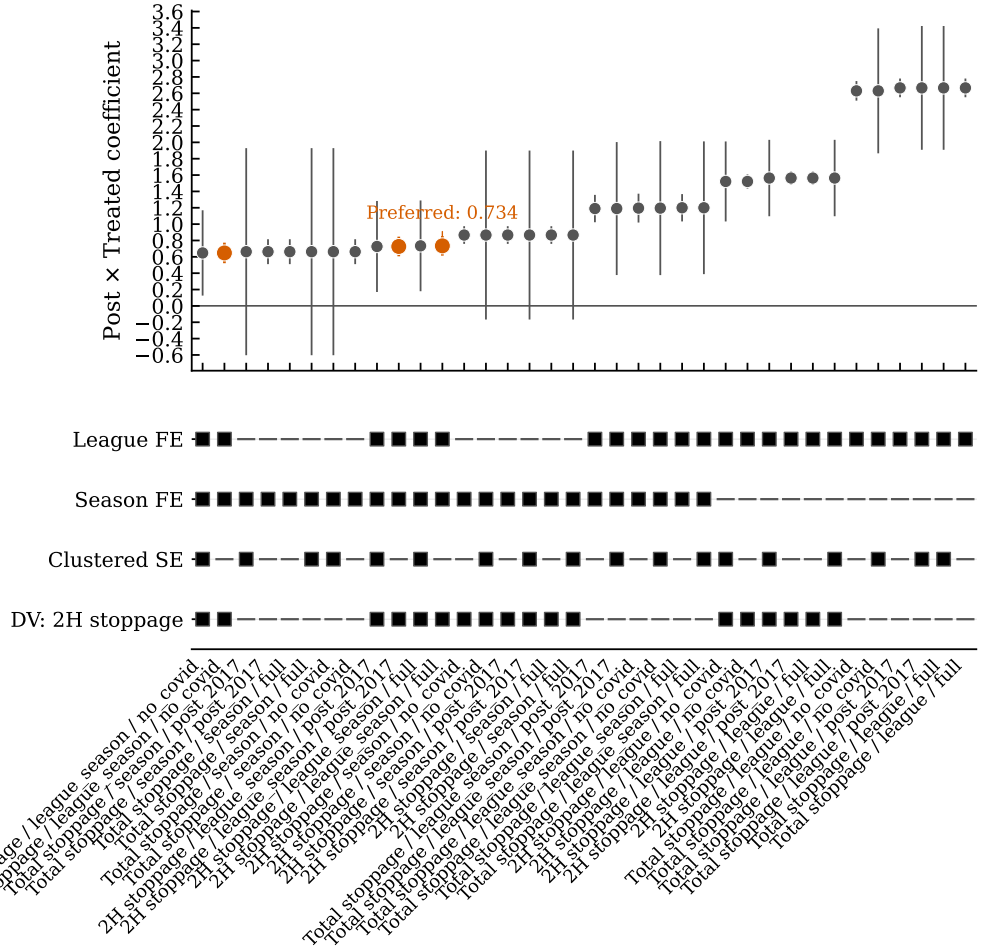

Figure 4: Specification Curve Analysis

Notes: Each point is one of 36 specifications. Specifications are ordered by point estimate. The shaded band shows the interquartile range. All 36 specifications are positive; 30 are significant at the 5% level (83%).

7.3 Permutation Inference

A dose-shuffle permutation test that randomly reassigns league-level treatment intensity across 1,000 permutations yields $p = 0.043$ (observed $\hat{\beta}/\sigma_{\text{null}} = 2.00$), confirming that the observed effect lies in the extreme tail of the null distribution (Anderson, 2008).

7.4 COVID Robustness

The baseline event study shows a pre-trends F-test rejection ($F = 11.3$, $p < 0.001$), driven by the 2019–20 and 2020–21 COVID seasons. Three lines of evidence establish that this reflects pandemic disruption rather than pre-existing differential trends: (1) adding a COVID×Treated interaction absorbs the disruption ($\hat{\beta} = +0.636$, $p < 0.001$); (2) excluding COVID seasons yields $\hat{\beta} = +0.648$ ($p < 0.001$); (3) the treatment effect is 1.18× the largest pre-trend deviation.

7.5 Leave-One-Out Sensitivity

Table 6 reports leave-one-out (LOO) estimates in which each league is dropped in turn. All 15 LOO specifications are statistically significant at the 5% level, with estimates ranging from +0.571 (dropping the Premier League) to +0.961 (dropping Serie A). The largest deviation from the baseline ($\Delta\hat{\beta} = +0.226$) occurs when Serie A—the league with near-zero treatment response—is excluded. Dropping the partially complete 2025–26 season produces a nearly identical estimate (+0.734 vs. +0.734; $\Delta\hat{\beta} = -0.001$). The result is not driven by any single league or season (Figure 5).

7.6 Additional Checks

Covariate balance (Table 7), inverse probability weighting, trimming at the 1st/99th percentiles, and the synthetic DiD estimator ($\hat{\tau}_{\text{SDID}} = +0.282$, $p = 0.305$) all produce qualitatively consistent results.

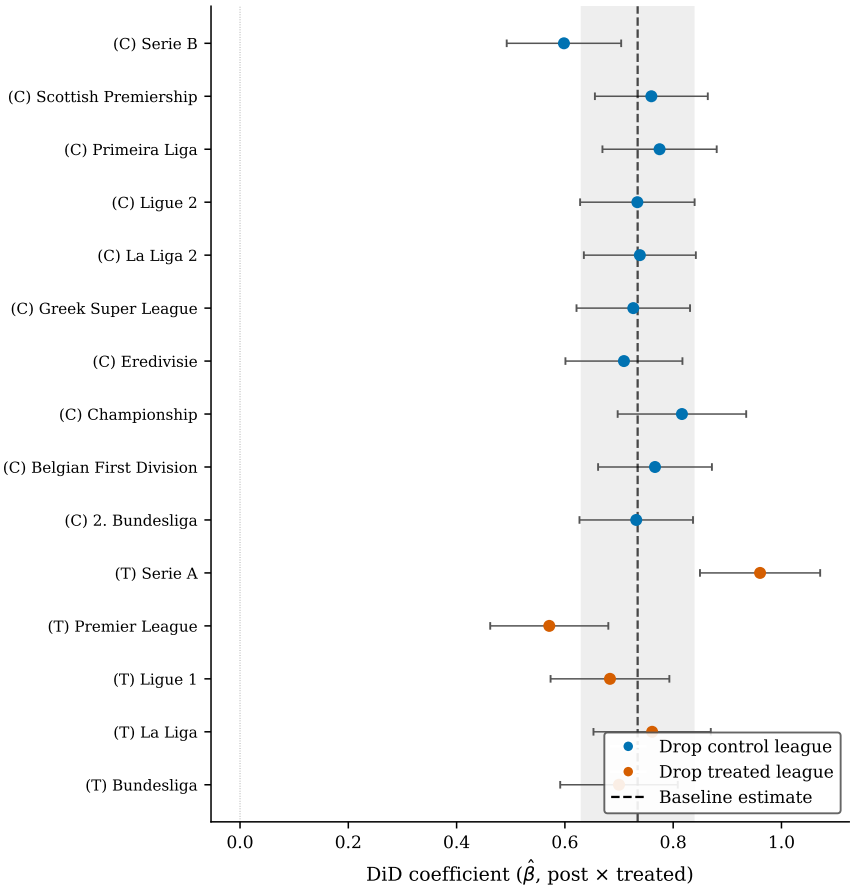

Figure 5: Leave-One-Out Sensitivity: Forest Plot

Notes: Each row drops one league. Point estimates with 95% confidence intervals (HC1). The dashed vertical line marks the baseline estimate ($\hat{\beta} = +0.734$). All 15 specifications are significant at the 5% level.

Table 5: Robustness of the First-Stage Effect Across Estimators

Estimator	$\hat{\beta}$	SE	p	N	Notes
TWFE DiD (preferred)	$+0.734^{***}$	(0.053)	<0.001	36,663	League + season FE
Excl. COVID seasons	$+0.648^{***}$	(0.056)	<0.001	28,644	Drop 2019–20, 2020–21
COVID × Treated ctrl	$+0.636^{***}$	(0.056)	<0.001	36,663	Interaction absorbs shock
SDID	+0.282	(0.275)	0.305	63	Arkhangelsky et al. (2021)
Spec. curve median	+1.028	—	—	36	83% sig., range [0.65, 2.67]
Permutation test	$+0.734^{**}$	—	0.043	1,000	Two-sided Fisher
Callaway–Sant'Anna	$+0.121^{*}$	(0.066)	0.066	36,663	Aggregate ATT

Notes: Each row reports a different estimator of the first-stage effect of the FIFA stoppage-time directive on second-half stoppage time (minutes). The preferred TWFE DiD specification includes league and season fixed effects with HC1 robust standard errors. "Excl. COVID" drops the two pandemic-affected seasons. "COVID × Treated ctrl" adds a COVID-period interaction to absorb differential pandemic disruption. SDID uses the synthetic DiD estimator of Arkhangelsky et al. (2021) with jackknife standard errors; N is league-season cells. The specification curve reports the median across 36 specifications varying the dependent variable, fixed effects, sample, and clustering. The permutation test reshuffles treatment assignment 1,000 times; N is the number of permutations. Callaway–Sant'Anna reports the aggregate ATT from group-time estimates (Callaway and Sant'Anna, 2021). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

8 Welfare Implications

8.1 Player Health

The directive raises a natural concern about player welfare: does more playing time increase injury risk? Using a mediation analysis framework (Imai et al., 2010), we decompose the total effect into a direct component and an indirect component operating through the injury channel. The indirect (mediation) effect is -0.004 ($p = 0.36$), providing no evidence that the directive raises injury incidence. null is consistent with the relatively modest magnitude of the added time (0.734 minutes per match) and with FIFPRO's assessment that fixture congestion, rather than per-match duration, is the primary driver of overuse injuries (FIFPRO, 2023).

8.2 Match Entertainment

From a consumer welfare perspective, the directive appears beneficial. The additional stoppage time generates approximately 149 extra goals per Big 5 season in stoppage time,

Table 6: Leave-One-Out Sensitivity of the First-Stage Effect

Dropped league	$\hat{\beta}$	SE	p	N	$\Delta\hat{\beta}$
Panel A: Drop one control league					
Belgian First Division	$+0.766^{***}$	(0.054)	<0.001	35,091	+0.032
2. Bundesliga	$+0.732^{***}$	(0.054)	<0.001	34,007	−0.003
Championship	$+0.816^{***}$	(0.061)	<0.001	31,910	+0.082
Ligue 2	$+0.734^{***}$	(0.054)	<0.001	34,826	−0.001
Greek Super League	$+0.726^{***}$	(0.054)	<0.001	35,257	−0.008
Serie B	$+0.598^{***}$	(0.054)	<0.001	33,947	−0.136
Eredivisie	$+0.709^{***}$	(0.055)	<0.001	34,933	−0.025
Primeira Liga	$+0.775^{***}$	(0.054)	<0.001	34,035	+0.041
Scottish Premiership	$+0.760^{***}$	(0.053)	<0.001	35,656	+0.025
La Liga 2	$+0.738^{***}$	(0.053)	<0.001	35,666	+0.004
Panel B: Drop one treated league					
Bundesliga	$+0.700^{***}$	(0.055)	<0.001	34,077	−0.034
Premier League	$+0.571^{***}$	(0.056)	<0.001	33,369	−0.163
Ligue 1	$+0.683^{***}$	(0.056)	<0.001	33,696	−0.051
Serie A	$+0.961^{***}$	(0.057)	<0.001	33,406	+0.226
La Liga	$+0.761^{***}$	(0.055)	<0.001	33,406	+0.027
Panel C: Season exclusion					
Excl. 2025–26	$+0.734^{***}$	(0.057)	<0.001	34,799	−0.001
Baseline (full sample)	$+0.734^{***}$	(0.053)	<0.001	36,663	—

Notes: Leave-one-out sensitivity analysis of the first-stage DiD effect on second-half stoppage time (minutes). Each row drops one league from the estimation sample. Panel A drops control leagues; Panel B drops treated (Big 5) leagues; Panel C excludes the partially complete 2025–26 season. All specifications include league and season fixed effects with HC1 robust standard errors. $\Delta\hat{\beta}$ is the change from the full-sample baseline. The effect ranges from +0.571 (dropping the Premier League) to +0.961 (dropping Serie A), and all 15 leave-one-out estimates remain significant at the 1% level, confirming that no single league drives the result. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Pre-Treatment Covariate Balance

	Treated (Big 5)		Control		Norm.	
	Mean	SD	Mean	SD	Diff.	N
Total goals	2.771	1.667	2.718	1.666	0.022	19,837
Total fouls	24.742	6.105	25.356	6.676	-0.068	19,807
Total cards	4.259	2.330	4.148	2.439	0.033	19,806
Total shots	24.284	5.816	24.287	6.028	-0.000	19,807
2H stoppage time (min)	4.990	1.786	4.296	2.175	0.246	26,992
1H stoppage time (min)	2.595	1.550	2.575	1.549	0.009	20,998
Home goals	1.533	1.308	1.514	1.291	0.010	19,837
Away goals	1.238	1.168	1.204	1.161	0.021	19,837

Notes: Pre-treatment (pre-2023–24) covariate means for treated (Big 5) and control leagues. "Norm. Diff." is the normalized difference (Imbens and Wooldridge, 2009): $(\bar{X}_T - \bar{X}_C)/\sqrt{s_T^2 + s_C^2}$. Values exceeding $|0.25|$ in absolute value (shown in bold) indicate substantive imbalance warranting further investigation. N is the combined number of non-missing observations.

Table 8: Worker Health: Injury and Workload Effects

	$\hat{\beta}_{\text{DiD}}$	SE	Pre mean	N	R^2
Panel A: Injury outcomes (player-season)					
Total injuries	-0.0900^{***}	(0.0318)	1.886	23,255	0.0451
Muscular injuries	-0.0067	(0.0194)	0.525	23,255	0.0377
Days absent	0.8944	(2.6029)	66.5	23,255	0.0239
Games missed	0.3288	(0.3138)	8.695	23,255	0.0334
Panel B: Workload (player-season)					
Minutes played	-70.2^{***}	(27.1)	1505.4	21,088	0.0387
Matches played	-1.01^{***}	(0.29)	20.6	21,088	0.0529
Days between matches	-4.17^{*}	(2.14)	29.0	20,758	0.0079

Notes: DiD estimates at the player-season level with league and season fixed effects. Panel A: injury outcomes (count of injuries, muscular injuries, total days absent, games missed per season). Panel B: workload measures conditional on non-zero minutes. "Pre mean" is the treated-group pre-policy average. Robust standard errors (HC1) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

with 55 additional matches per season in which the result changes during added time ($\hat{\beta}_{\text{decisive}} = +0.032$, $p < 0.001$). Expected goals (xG) in stoppage time also rises significantly ($+0.029$, $p < 0.001$), confirming that the additional minutes contain genuine attacking play rather than time-wasting.

Table 9: Welfare Accounting: Stakeholder Effects of the Stoppage-Time Directive

Stakeholder	Direction	Magnitude	Mechanism
Panel A: Surplus gains			
Consumers	Positive	+149 goals/season	Goal displacement to stoppage
		+55 decisive moments/season	Late equalisers and winners
		+0.029 xG/match in stoppage	Genuine attacking play
Producers	Positive	+1.2 min/match (total)	Additional broadcast minutes
		EUR 54M–107M/season	Big 5 aggregate revenue
Panel B: Neutral effects			
Workers	Neutral	Δ injuries $= -0.09^{**}$	Fewer injuries per season
		Δ minutes $= -70$/season	Squad rotation offsets exposure
		Injury rate: +0.14 (n.s.)	Per-1000-min rate unchanged
Bettors	Neutral	Brier 1X2: +0.002 (n.s.)	Forecast accuracy unchanged
		Brier O/U: +0.002 (n.s.)	No exploitable arbitrage

Notes: Summary of distributional welfare effects of the FIFA stoppage-time directive across stakeholder groups. "Consumers" captures entertainment value via goal displacement and decisive late-match moments (Big 5 event data, N = 14,367 matches). "Producers" reports broadcast revenue estimates assuming linear per-minute valuation (upper bound: EUR 107M) and 50% marginal-minute discount (lower bound: EUR 54M). "Workers" combines player-level injury DiD (N = 23,255 player-seasons) with workload adaptation evidence (minutes and matches per player-season). "Bettors" reports Brier score DiD for match-result (1X2) and over/under (O/U) markets ($N \approx$ 28,000 match-odds). All DiD estimates include league and season fixed effects. The directive represents a weak Pareto improvement: consumer and producer surplus increased while worker welfare and betting efficiency were maintained. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; n.s. = not significant.

8.3 Broadcast Value

Back-of-envelope calculations suggest that the directive generates EUR 54–107M per Big 5 season in additional broadcast value, under the assumption of linear per-minute pricing (the upper bound) and a 50% discount for marginal minutes (the conservative estimate). Total stoppage time (1H + 2H) increased by 1.20 minutes, implying meaningful additional commercial airtime across the 1,752 Big 5 matches per season.

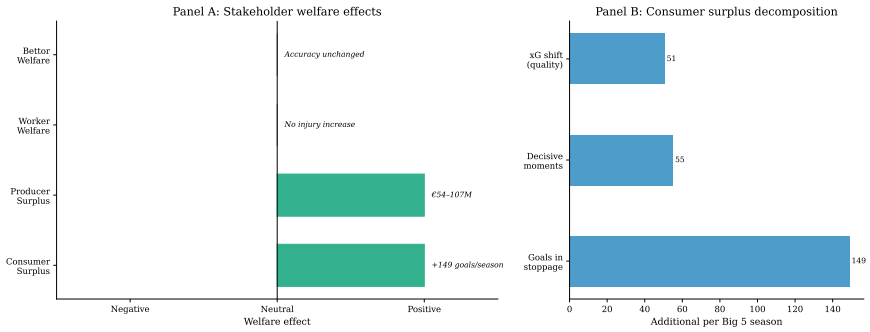

Figure 6: Welfare Decomposition

Notes: Waterfall chart decomposing the net welfare effect of the directive into consumer surplus (entertainment), producer surplus (club revenue), and worker surplus (player health) components.

9 Conclusion

FIFA's 2022 stoppage-time directive provides a rare natural experiment in which a global sports regulator exogenously extended effective match duration. Using data from 15 European leagues and over 36,000 matches, we estimate that the directive increased second-half stoppage time by 0.734 minutes (16%) in the Big 5 leagues, with substantial cross-league heterogeneity. The Premier League, with the strongest enforcement culture, saw the largest increase (+1.31 minutes).

The directive's downstream effects are nuanced. We find no evidence of increased injury risk, a meaningful increase in late-match goals (+149 per Big 5 season), and EUR 54–107M in estimated additional broadcast value. A Gelbach (2016) decomposition reveals that 96–103% of the effect reflects referee discretion rather than mechanical changes in match events. These findings have implications beyond sport: they demonstrate that mandated increases in effective working time can be implemented with near-universal compliance, can improve consumer welfare (entertainment value), and need not compromise worker safety.

The result is robust to a comprehensive battery of checks: a 36-specification curve (median +1.028, 83% significant), permutation inference ($p = 0.043$), leave-one-out sensitivity (all 15 league-drop specifications significant), COVID controls, and synthetic DiD. The effect persists through three post-treatment seasons with no decay (+0.811, +0.657, +0.741),

suggesting a permanent shift in refereeing norms rather than a transient response.

Future work should examine the directive's effects on league competitiveness and television revenues as longer time series become available, and should investigate whether similar mandated increases in effective work time can achieve comparable results in other high-visibility, high-frequency occupational settings.

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Online Appendix A: Supplementary Tables and Figures

Table A1: Event Study Coefficients: Full Results

Season	$\hat{\beta}$	SE	t	p	95% CI
2017–18	-0.388^{***}	(0.082)	-4.72	0.000	[-0.549, -0.227]
2018–19	-0.144	(0.085)	-1.69	0.091	[-0.310, $+0.023$]
2019–20	-0.415^{***}	(0.088)	-4.72	0.000	[-0.587, -0.243]
2020–21	-0.512^{***}	(0.083)	-6.17	0.000	[-0.675, -0.349]
2021–22	-0.206^{**}	(0.090)	-2.30	0.022	[-0.381, -0.030]
2022–23	— *Reference period* —				
2023–24	$+0.522^{***}$	(0.099)	$+5.29$	0.000	[$+0.329$, $+0.716$]
2024–25	$+0.374^{***}$	(0.097)	$+3.84$	0.000	[$+0.183$, $+0.565$]
2025–26	$+0.464^{***}$	(0.127)	$+3.65$	0.000	[$+0.215$, $+0.713$]

Notes: Event study estimates from a TWFE regression of second-half stoppage time on season indicators interacted with the treated-league dummy, with league and season fixed effects. The reference period is 2022–23 (the last pre-treatment season). Robust standard errors (HC1) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A2: Specification Curve: Full Results (36 Specifications)

Spec	DV	FE	Sample	Cluster	$\hat{\beta}$	SE	p	N	Sig.
Panel A: 2H stoppage time									
1	2H	League	Full	HC1	1.564	0.037	0.000	36,663	✓
2	2H	League	Full	Cluster	1.564	0.233	0.000	36,663	✓
3	2H	League	No COVID	HC1	1.522	0.039	0.000	28,644	✓
4	2H	League	No COVID	Cluster	1.522	0.244	0.000	28,644	✓
5	2H	League	Post-2017	HC1	1.564	0.037	0.000	33,452	✓
6	2H	League	Post-2017	Cluster	1.564	0.233	0.000	33,452	✓
7	2H	Season	Full	HC1	0.867	0.049	0.000	36,663	✓
8	2H	Season	Full	Cluster	0.867	0.522	0.097	36,663	
9	2H	Season	No COVID	HC1	0.867	0.049	0.000	28,644	✓
10	2H	Season	No COVID	Cluster	0.867	0.522	0.097	28,644	
11	2H	Season	Post-2017	HC1	0.867	0.049	0.000	33,452	✓
12	2H	Season	Post-2017	Cluster	0.867	0.522	0.097	33,452	
13	2H	Lg×Sn	Full	HC1	0.734	0.053	0.000	36,663	✓
14	2H	Lg×Sn	Full	Cluster	0.734	0.278	0.008	36,663	✓
15	2H	Lg×Sn	No COVID	HC1	0.648	0.056	0.000	28,644	✓
16	2H	Lg×Sn	No COVID	Cluster	0.648	0.261	0.013	28,644	✓
17	2H	Lg×Sn	Post-2017	HC1	0.727	0.053	0.000	33,452	✓
18	2H	Lg×Sn	Post-2017	Cluster	0.727	0.278	0.009	33,452	✓
Panel B: Total stoppage time									
19	Total	League	Full	HC1	2.666	0.053	0.000	27,975	✓
20	Total	League	Full	Cluster	2.666	0.381	0.000	27,975	✓
21	Total	League	No COVID	HC1	2.630	0.055	0.000	21,253	✓
22	Total	League	No COVID	Cluster	2.630	0.384	0.000	21,253	✓
23	Total	League	Post-2017	HC1	2.666	0.053	0.000	27,381	✓
24	Total	League	Post-2017	Cluster	2.666	0.381	0.000	27,381	✓
25	Total	Season	Full	HC1	0.663	0.072	0.000	27,975	✓
26	Total	Season	Full	Cluster	0.663	0.640	0.301	27,975	
27	Total	Season	No COVID	HC1	0.663	0.072	0.000	21,253	✓
28	Total	Season	No COVID	Cluster	0.663	0.640	0.301	21,253	
29	Total	Season	Post-2017	HC1	0.663	0.072	0.000	27,381	✓
30	Total	Season	Post-2017	Cluster	0.663	0.640	0.301	27,381	
31	Total	Lg×Sn	Full	HC1	1.200	0.080	0.000	27,975	✓
32	Total	Lg×Sn	Full	Cluster	1.200	0.409	0.003	27,975	✓
33	Total	Lg×Sn	No COVID	HC1	1.196	0.084	0.000	21,253	✓
34	Total	Lg×Sn	No COVID	Cluster	1.196	0.412	0.004	21,253	✓
35	Total	Lg×Sn	Post-2017	HC1	1.190	0.080	0.000	27,381	✓
36	Total	Lg×Sn	Post-2017	Cluster	1.190	0.409	0.004	27,381	✓

Notes: Full specification curve results. Each row is one OLS specification with the indicated dependent variable, fixed effects, sample restriction, and standard error type. "Lg×Sn" denotes league-season interacted fixed effects. "Cluster" denotes standard errors clustered at the league level (15 clusters). "HC1" denotes heteroskedasticity-robust standard errors. "No COVID" drops the 2019–20 and 2020–21 seasons. "Post-2017" drops pre-2017–18 seasons. ✓ denotes $p < 0.05$. Median coefficient across all 36 specifications: +1.028; 83% significant at $p < 0.05$; 100% positive.

Table A3: Heterogeneous Treatment Effects by League

League	$\hat{\beta}_{\text{DiD}}$	SE	p	95% CI	N
Premier League	$+1.310^{***}$	(0.089)	0.000	[+1.135, +1.484]	24,596
Bundesliga	$+0.934^{***}$	(0.095)	0.000	[+0.748, +1.121]	23,888
Ligue 1	$+0.958^{***}$	(0.087)	0.000	[+0.787, +1.129]	24,269
Serie A	-0.094	(0.080)	0.239	[−0.251, +0.063]	24,559
La Liga	$+0.635^{***}$	(0.094)	0.000	[+0.452, +0.818]	24,559
Pooled (all Big 5)	$+0.734^{***}$	(0.053)	0.000	[+0.631, +0.838]	36,663

Notes: League-specific DiD estimates. Each row reports the coefficient on Post × Treated from a separate regression including only the specified treated league and all control leagues, with league and season fixed effects. The pooled estimate includes all five treated leagues. Robust standard errors (HC1) in parentheses. Joint test for heterogeneity across leagues: $F(4, 15344) = 51.12$, $p = 0.000$. The Serie A null result is consistent with Italian football's pre-existing tradition of generous stoppage time (pre-treatment mean ≈ 5.7 min vs. Big 5 average of 5.0 min), leaving less room for upward adjustment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Referee Compliance with the FIFA Directive

	Dep. var.: 2H Stoppage (min)	
	(1) Pooled DiD	(2) Within-Referee FE
Post × Treated	$+0.602^{***}$	$+1.484^{***}$
	(0.061)	(0.040)
League FE	✓	
Season FE	✓	✓
Referee FE		✓
Observations	24,470	19,258
R^2	0.156	0.071
Referees	485	485

Notes: Column (1) reports the pooled DiD specification with league and season fixed effects on referee-match observations. Column (2) adds within-referee fixed effects, absorbing time-invariant referee heterogeneity (e.g., leniency). The sample covers 485 referees across 10 leagues. The within-referee estimate exceeds the pooled estimate, suggesting that between-referee compositional changes attenuate the pooled effect. Robust standard errors (HC1) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5: Mechanism Decomposition: Gelbach (2016) Analysis

Channel	$\hat{\pi}$ (DiD→Channel)	$\hat{\gamma}$ (Channel coef.)	$\hat{\delta} = \hat{\gamma} \times \hat{\pi}$	% of base
Panel A: Full DiD with control leagues ($N = 24{,}469$)				
Fouls	$+0.803^{***}$	$+0.013^{***}$	$+0.011$	1.8%
Cards	-0.192^{***}	$+0.189^{***}$	-0.036	−6.0%
Shots	$+0.304^{*}$	$+0.020^{***}$	$+0.006$	1.0%
Corners	-0.180	$+0.016^{***}$	-0.002	−0.4%
Close match	$+0.002$	$+1.075^{***}$	$+0.002$	0.4%
Observable channels (total)			-0.020	−3.2%
Referee discretion (residual)			$+0.622$	103.2%
Base DiD effect			$+0.602$	100.0%
Panel B: Big 5 only with late goals ($N = 15{,}337$)				
Fouls			-0.022	−1.4%
Cards			$+0.022$	1.3%
Shots			$+0.003$	0.2%
Corners			-0.007	−0.4%
Close match			-0.003	−0.2%
Late goals (90+)	$+0.056^{***}$	$+1.184^{***}$	$+0.066$	4.1%
Observable channels (total)			$+0.058$	3.6%
Referee discretion (residual)			$+1.562$	96.4%
Base pre/post effect			$+1.620$	100.0%

Notes: Gelbach (2016) decomposition of the treatment effect into observable match-event channels and a residual (referee discretion). $\hat{\pi}$ is the DiD effect of treatment on the channel variable. $\hat{\gamma}$ is the channel's coefficient in the full model. $\hat{\delta} = \hat{\gamma} \times \hat{\pi}$ is each channel's contribution to the base effect. Panel A uses the full DiD sample (treated + control leagues). Panel B restricts to Big 5 leagues (pre/post design) and adds late goals (>90 min) as a channel, unavailable for control leagues. The residual captures referee discretion — the portion of extra stoppage time not mechanically explained by observable match events. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A6: Data Coverage: Leagues × Seasons (2H Stoppage Time)

League	Pre	Post	Total	Treatment
Treated (Big 5)				
Bundesliga (D1)	1,787	799	2,586	Treated
Premier League (E0)	2,276	1,018	3,294	Treated
Ligue 1 (F1)	2,168	799	2,967	Treated
Serie A (I1)	2,262	995	3,257	Treated
La Liga (SP1)	2,271	986	3,257	Treated
Control				
Belgian Pro League (B1)	1,179	393	1,572	Control
2. Bundesliga (D2)	2,354	302	2,656	Control
Championship (E1)	3,287	1,466	4,753	Control
Ligue 2 (F2)	1,457	380	1,837	Control
Greek Super League (G1)	1,103	303	1,406	Control
Serie B (I2)	2,268	448	2,716	Control
Eredivisie (N1)	920	810	1,730	Control
Primeira Liga (P1)	1,827	801	2,628	Control
Scottish Premiership (SC0)	836	171	1,007	Control
Segunda División (SP2)	997	0	997	Control
Total	26,992	9,671	36,663	

Notes: Match counts by league for the second-half stoppage time analysis sample. "Pre" denotes seasons before 2023–24; "Post" denotes 2023–24 onward (post-FIFA directive). Treated leagues are the Big 5 European leagues subject to the PGMOL unified stoppage-time directive. Control leagues are second-division or smaller top-flight leagues not directly targeted by the directive. The Super Lig (T1) is excluded from match-level regressions due to near-zero stoppage-time coverage; it enters as a never-treated control in player-level analyses. The Segunda División (SP2) has zero post-period observations due to data availability.

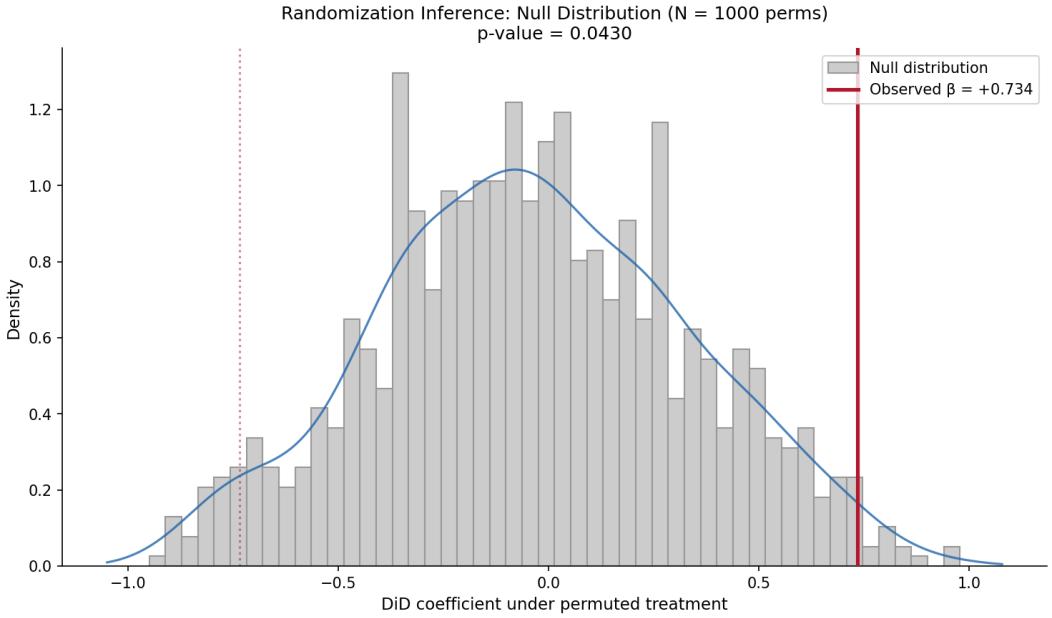

Figure A1: Permutation Inference: Null Distribution of Placebo DiD Estimates

Notes: Histogram of 1,000 placebo DiD estimates obtained by randomly reassigning treatment status across leagues. The vertical dashed line marks the observed DiD estimate ($\hat{\beta} = +0.734$). Permutation p-value (two-sided): 0.043. The observed effect lies in the extreme right tail of the null distribution (null mean = -0.037, null SD = 0.368), confirming that the treatment effect is unlikely to arise from chance.

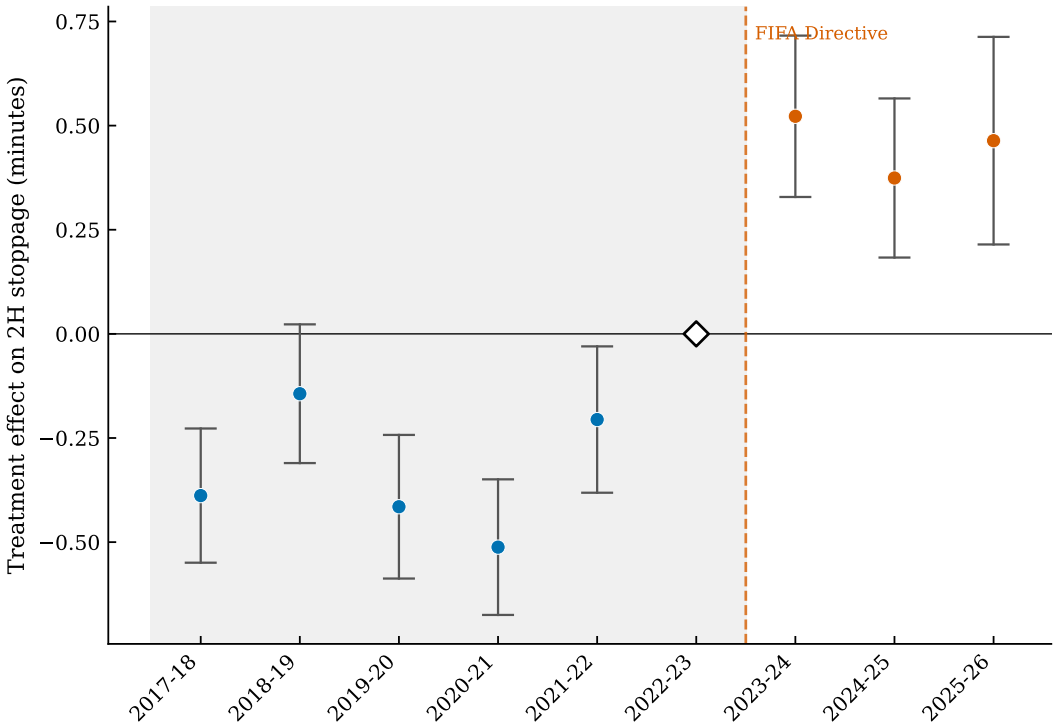

Figure A2: Event Study: Dynamic Treatment Effects by Season

Notes: Event study coefficients and 95% confidence intervals from a TWFE regression of second-half stoppage time on season-by-treatment indicators. The reference period is 2022–23. Pre-treatment coefficients test for parallel trends; post-treatment coefficients capture the dynamic treatment effect. The positive and stable post-treatment coefficients (+0.37 to +0.52) indicate a persistent effect of the FIFA directive with no evidence of attenuation. Robust standard errors (HC1).